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EXP - 5

TITLE - Sum of Subset Problem

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In [2]: def Sum_of_Subset(S, d):
        result = []

        def backtrack(i, current_subset, current_sum):
            if current_sum == d:
                result.append(current_subset)
                return
            if i == len(S) or current_sum > d:
                return

            backtrack(i + 1, current_subset + [S[i]], current_sum + S[i])
            backtrack(i + 1, current_subset, current_sum)

        backtrack(0, [], 0)
        return result

S = list(map(int, input("Enter the elements of set S: ").split()))
d = int(input("Enter the sum d:"))

subsets = Sum_of_Subset(S, d)
print("=====")
if subsets:
    print("Subsets whose sum is", d, ":")
    for subset in subsets:
        print(subset)
else:
    print("No subset found with sum", d)
print("=====")
```

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=====
Subsets whose sum is 60 :
[10, 20, 30]
[10, 50]
[20, 40]
[60]
=====
```

In []: